

Sequence Diagram: Check Out Book

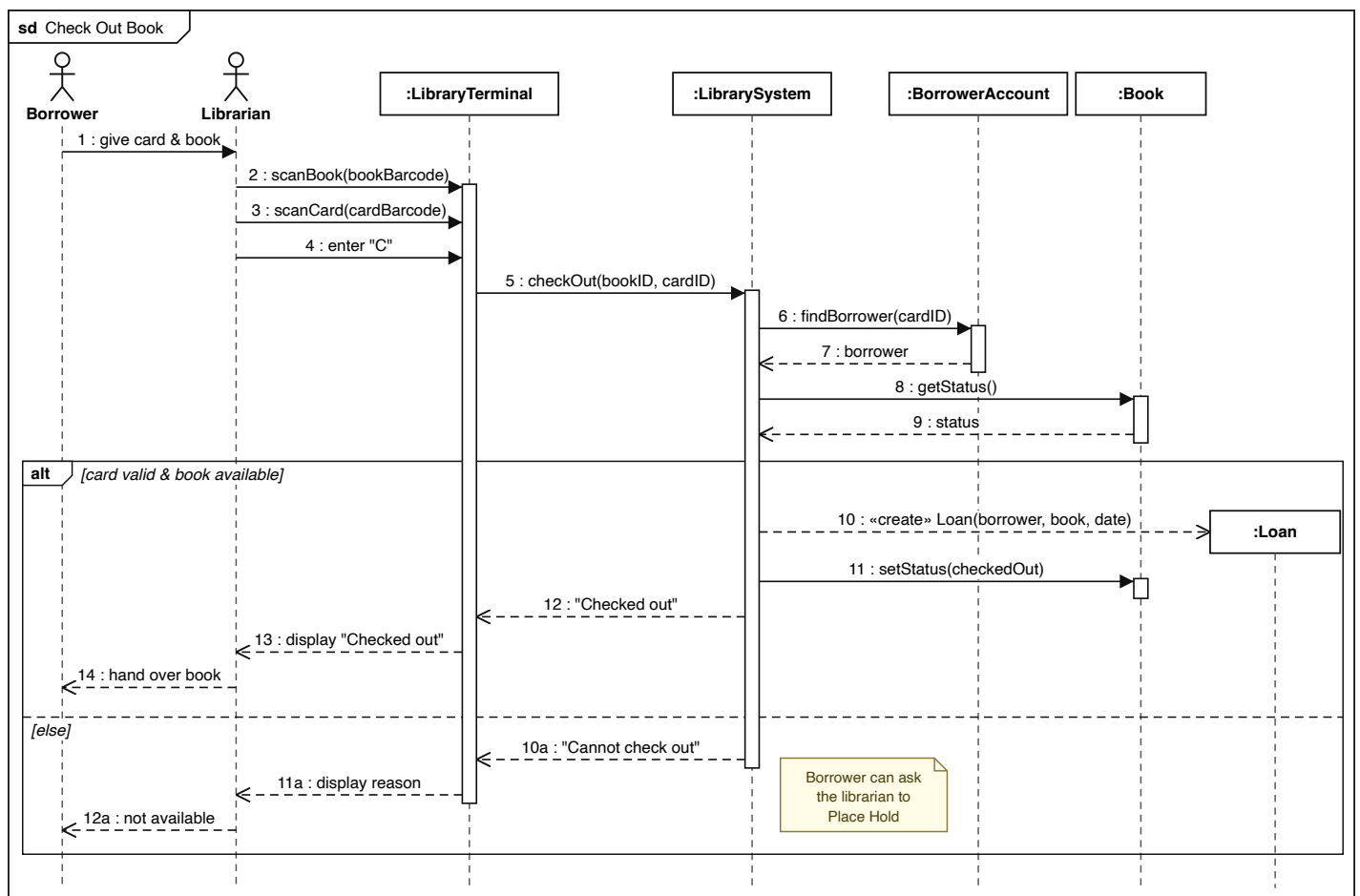
CS 200, Team 37 · Faizan Khan, Maggie Skinner, Martin Devanathan, Tallon King, Alyssa Duncan

Selected Use Case

We chose **Check Out Book** from the Library System. Actors: **Librarian** (works the terminal) and **Borrower**. Every book and every borrower card has a bar code. When a borrower wants to check out a book, the librarian scans the bar code on the book and on the borrower's card, then enters **C** at the terminal. The system finds the borrower's account and checks the book's status. If the card is valid and the book is available, it creates a loan and marks the book as checked out. If the book is already checked out, the check-out is refused and the borrower can ask the librarian to place a hold (the **Place Hold** use case).

Related Classes

Borrower (actor)	Person who wants to check out a book; has a card with a bar code.	BorrowerAccount	The borrower's record, found by the card's bar code.
Librarian (actor)	Scans bar codes and enters commands at the terminal.	Book	A book in the collection, with a bar code and a status (available, checked out, on hold).
LibraryTerminal	Boundary class: bar code scanner, keyboard and screen.	Loan	Created at check-out; links one borrower to one book with the check-out date.
LibrarySystem	Control class: handles the check-out command and coordinates the other objects.		



Solid arrows with filled heads are calls and dashed arrows are returns. Step 4 is the librarian entering the check-out code C. The **alt** fragment shows both outcomes: a loan is created when the card is valid and the book is available (10–14); otherwise no loan is created and the librarian tells the borrower why (10a–12a).